

2 - Intelligent Agents

Artificial Intelligence Lab Manual - Clean Study Edition

Learning Objectives

- Describe an agent in terms of percepts, actions, state, and environment.
- Distinguish simple reflex, model-based, goal-based, utility-based, and learning agents.
- Implement a small rule-driven agent and evaluate its behavior.

Background

An intelligent agent observes an environment and selects actions. The key design question is not whether the code looks sophisticated, but whether the chosen representation and decision rule allow the agent to act effectively under the available information.

Core Concepts

Concept	Meaning in this lab
Percept	Information received by the agent at a particular moment.
State	Internal representation used to summarize relevant history.
Action	An operation the agent can perform in the environment.
Policy / rule	Mapping from percept or state to an action.
Performance measure	Criterion used to judge whether the agent is behaving well.

Guided Procedure

1. Define a small environment such as a two-room vacuum world or a temperature-control system.
2. List possible percepts, actions, and the performance measure.
3. Implement a simple reflex agent with explicit condition-action rules.
4. Add a small internal state so the agent can remember one relevant fact from an earlier step.
5. Run at least three scenarios and compare the reflex and stateful versions.

Reference Implementation

Use the following as a starting point. Read and modify it rather than treating it as a final answer.

```
def vacuum_agent(location, dirty):
    if dirty:
        return "CLEAN"
    if location == "A":
        return "MOVE_RIGHT"
    return "MOVE_LEFT"

scenarios = [("A", True), ("A", False), ("B", False)]
for s in scenarios:
    print(s, "->", vacuum_agent(*s))
```

Lab Exercises

- Task 1.** Design a traffic-light agent using current queue lengths as percepts. State the performance measure.
- Task 2.** Modify the vacuum agent so it remembers whether the other room was already observed clean.
- Task 3.** Give one example where a simple reflex agent fails because the current percept is insufficient.

Task 4. For a delivery robot, classify the environment as fully/partially observable, deterministic/stochastic, and static/dynamic.

Suggested Deliverables

- Notebook or Python file containing the completed implementation.
- Short result table or output evidence for each required comparison.
- A concise interpretation of what changed and why; do not submit output without explanation.

Review Questions

1. What makes a rational action different from a guaranteed successful action?
2. When is internal state useful?
3. How does a utility-based agent differ from a goal-based agent?

Completion check: You should be able to explain the algorithm or workflow in your own words, reproduce the main result, and identify at least one limitation.